

Exact Affine-Transfer Reformulations of Collatz

Collatz proof project — September 5, 2026

Let $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n . Write $\mathcal{B}(n)$ for boundedness of its natural orbit and $\mathcal{R}(n)$ for the existence of $t \geq 0$ with $T^t(n) = 1$. The previously proposed transfer $z \mapsto 9z + 2$ is not established here. We determine its exact logical strength, including the effect of restricting its domain.

Theorem 1 (Nondivergence). *The following three assertions are equivalent:*

$$\begin{aligned} \forall n \geq 0, \mathcal{B}(n); \\ \forall z \geq 0, \mathcal{B}(z) \implies \mathcal{B}(9z + 2); \\ \forall z \equiv 2 \pmod{3}, \mathcal{B}(z) \implies \mathcal{B}(9z + 2). \end{aligned}$$

Moreover, any unbounded natural orbit implies the existence of a $z \equiv 2 \pmod{3}$ with $\mathcal{B}(z)$ and $\neg\mathcal{B}(9z + 2)$.

Theorem 2 (Full conjecture). *The Collatz conjecture is equivalent to restricted reaching-one transfer:*

$$(\forall n > 0, \mathcal{R}(n)) \iff (\forall z \equiv 2 \pmod{3}, \mathcal{R}(z) \implies \mathcal{R}(9z + 2)).$$

This version includes the exclusion of nontrivial cycles. Neither side is proved by the equivalence.

Common argument. Use either a least unbounded seed, or strong induction on a seed whose convergence is to be established. Zero is bounded; one reaches one. Every positive even seed has a smaller positive successor. For an odd $n > 1$, write $n + 1 = 2^a m$ with m odd. If $a = 1$, two steps give $(3n + 1)/4 < n$. Thus only $a \geq 2$ remains.

The classical odd-run identities, formalized in `OddRunMerges`, give two cases at time $a + 2$. If $3^a m \equiv 1 \pmod{4}$, the seeds $n, n - 1$ merge. The smaller seed supplies boundedness or convergence of the common tail. Otherwise, putting $z = T^{a+2}(n - 1)$ gives

$$T^{a+2}(n) = 9z + 2, \quad 4z + 1 = 3^{a-1}m.$$

Since $a \geq 2$, the second identity gives $z \equiv 2 \pmod{3}$. The positive seed $n - 1$ supplies $\mathcal{B}(z)$ or $\mathcal{R}(z)$; the restricted transfer supplies the corresponding property of $T^{a+2}(n)$ and hence of n .

For boundedness, if any orbit were unbounded, its least seed would therefore produce the asserted failure pair $z, 9z + 2$. For convergence, this reasoning is a strong-induction proof using the transfer as a hypothesis. Reaching one is invariant under finite shifts, including shifts past an earlier hit of one: the orbit then stays in the cycle 1, 2. Conversely, universal boundedness or universal convergence immediately implies its corresponding transfer statement. $\square$

Formal evidence. `Collatz.AffineBoundedness` defines `OrbitBounded`, constructs the bounded/unbounded failure pair, and proves both boundedness equivalences. `Collatz.AffineConvergence` defines `ReachesOne`, proves shift invariance, and proves the full-conjecture equivalence by strong induction. The supporting residue theorem is in `Collatz.OddRunMerges`. All eight added or newly selected declarations are included in the axiom audit. No numerical search supplies an unproved premise.

Interpretation. Restricting transfer to $z = 2$ modulo three does not make it a consequence of the existing local identities. It still carries the entire target statement. In the reaching-one version the equivalent form is $\mathcal{R}(3u + 2) \Rightarrow \mathcal{R}(27u + 20)$ for every natural u . A proof must establish that implication; testing many convergent pairs or merely observing an affine relationship is insufficient. This note is an exact reformulation and a check against circular reasoning, not a proof of Collatz or a claim of mathematical priority.

Reproduction and background. From `lean/`, build `Collatz.AffineBoundedness` and `Collatz.AffineConvergence`; from the root run the theorem audit with `--build`. The classical endpoint identities used above are also in M. D. LaDue, *Clusters of Integers with Equal Total Stopping Times in the $3x + 1$ Problem* (2017), Lemma 3.1 and Theorem 4.1, arXiv:1709.02979.